hymeko_sbsh: the orthtree generic over its coordinate type

HyMeKo graph

Created-at: 2026-09-24 12:40 CEST

Approval. This is a core item (`CORE.YAML`: `hymeko_sbsh`). The user directed it on 2026-09-24: "Let's make the hymeko_sbsh generic templates so the types can be set easily".

Scope. The design has three parts:

- a `Coordinate` trait, implemented for `f32` and `f64`, providing ordering, `+`, `-`, `*`, `ZERO`, `HALF`, `min` and `max`;
- `OrthTree<T = f64>`, `SplitRule<T = f64>`, `build_orthtree<T>` and `point_bbox<T>`;
- `CountSplit` implements the rule for every `T`.

The arithmetic is unchanged (the midpoint is still $lo \cdot \frac{1}{2} + hi \cdot \frac{1}{2}$). The `f64` default keeps every caller source-compatible: the scene, `BoundsIndex` and the tests. No behaviour change for `f64`.

Tests.

- The existing suites pass.
- New: an `f32` tree over exactly representable coordinates equals the `f64` tree, in cells, levels, parents, children, members, boxes and `leaf_of`, for dim = 1..3.
- New: a generic caller rule.

Not in scope. Nagare's own f32 orthtree differs in more than precision: child bit order, integer-lattice midpoints, the degenerate-split rule, `leaf_of` on a dropped orthant, and the bounding-box padding. Migrating it is a separate decision.

Rollback. Revert the branch.

